

Environmental Science and Sustainable Engineering

CO900G

Alternative Assessment Method (AAM)

INDIVIDUAL TECHNICAL REPORT

E-WASTE RADAR

A Smart Web-Based Platform for E-Waste Collection Facility Discovery

Environmental Problem: Improper Disposal of Electronic Waste

Primary SDG: Sustainable Development Goal 12 – Responsible Consumption and Production

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ABSTRACT

Electronic waste, commonly known as e-waste, is one of the fastest-growing waste streams in the world. Electronic devices such as mobile phones, laptops, chargers, batteries and other appliances are frequently discarded when they become outdated or unusable.

Improper disposal can result in environmental pollution and the loss of valuable recoverable materials. One important reason for improper disposal is the difficulty people face in finding appropriate collection facilities. Although recycling and collection facilities may exist, users may not know where they are located, what types of e-waste they accept, whether they are currently open, or how to contact them.

This report proposes E-Waste Radar, a web-based platform designed to help users locate suitable e-waste collection facilities. The system allows users to select the type of electronic waste they want to dispose of and enter their location. It then displays suitable collection facilities based on factors such as distance, accepted waste categories and operating status. The proposed system also provides collection-drive information and basic preparation instructions before disposal, such as backing up personal data and removing SIM or memory cards. The solution uses web technologies such as HTML, CSS and JavaScript along with a database and map-based location services.

E-Waste Radar aims to make responsible e-waste disposal more convenient and accessible while supporting the objectives of sustainable waste management.

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INTRODUCTION

The rapid development of technology has increased the use of electronic devices in everyday life. Smartphones, computers, tablets, televisions, chargers and other electronic products are replaced frequently because of technological advancements, damage or changing consumer preferences. As a result, the amount of electronic waste generated worldwide continues to increase. According to the Global E-waste Monitor 2024, approximately 62 billion kg of e-waste was generated globally in 2022, while only about 22.3% was documented as formally collected and recycled. The report also projects that global e-waste generation could reach approximately 82 billion kg by 2030 if current trends continue.

E-waste is different from ordinary household waste because it can contain both valuable materials and potentially hazardous substances. Materials such as copper, aluminium, gold and other metals can potentially be recovered and reused. At the same time, some electronic products contain substances that can cause environmental and health concerns when improperly handled.

In India, e-waste management is regulated under the E-Waste (Management) Rules, 2022, which came into effect on 1 April 2023. The regulations emphasize responsible management and the role of producers, recyclers and other stakeholders in the e-waste management system.

However, regulations alone cannot solve the problem. Users also need simple ways to participate in responsible disposal. A common situation is that a person knows that an old laptop or phone should not simply be thrown into regular waste, but does not know where to take it. Searching for suitable facilities manually can be inconvenient, especially when different facilities accept different categories of e-waste.

This creates an accessibility gap between knowing that e-waste should be recycled and actually being able to dispose of it responsibly. E-Waste Radar is proposed to address this gap through a simple location-based web platform.

PROBLEM STATEMENT

Improper disposal of electronic waste is a significant environmental problem. Many users dispose of unwanted electronic devices through general waste channels because they lack convenient information about appropriate collection and recycling facilities. The main problems identified are:

1. Difficulty in locating nearby e-waste collection facilities.
2. Lack of information about which types of e-waste a facility accepts.
3. Uncertainty about operating hours and availability.
4. Difficulty finding temporary collection drives.
5. Lack of awareness about safe preparation of electronic devices before disposal.
6. Dependence on scattered information from different websites or sources.

Existing information may require users to search through multiple sources before finding a suitable facility. Therefore, there is a need for a simple digital platform that connects users with appropriate e-waste collection points based on location and waste category.

Problem Definition

“To develop a web-based system that helps users quickly identify and access suitable e-waste collection facilities based on their location and type of electronic waste.”

The proposed solution must be different from systems that simply classify objects as reusable or recyclable and should not depend on loyalty or reward points.

SUSTAINABILITY ANALYSIS

Relevant Sustainable Development Goals

SDG 12: Responsible Consumption and Production

SDG 12 is the primary Sustainable Development Goal associated with the proposed system. Target 12.5 aims to substantially reduce waste generation through prevention, reduction, recycling and reuse by 2030. E-Waste Radar supports this target by making responsible collection and recycling easier for users. The system also supports Target 12.8, which focuses on ensuring that people have relevant information and awareness for sustainable development and lifestyles.

The platform provides users with information about:

1. Appropriate collection facilities.
2. Accepted categories of e-waste.
3. Collection drives.
4. Disposal preparation.
5. Responsible disposal practices.

SDG 11: Sustainable Cities and Communities

Improper waste disposal affects urban environments. E-Waste Radar can contribute to more sustainable communities by helping residents connect with appropriate waste-management infrastructure.

SDG 9: Industry, Innovation and Infrastructure

The proposed system uses digital technology to improve access to environmental infrastructure. It demonstrates how information technology can be applied to a practical sustainability problem.

ISO 14001

ISO 14001 provides a framework for an Environmental Management System (EMS). The standard focuses on improving environmental performance, meeting applicable requirements and managing environmental risks.

E-Waste Radar does not itself make an organization ISO 14001 certified. Instead, the principles of environmental management can be applied to the proposed system.

The platform supports:

1. Better control of waste-related information.
2. Responsible waste handling.
3. Improved awareness.
4. Monitoring of collection activities.
5. Continuous improvement through collected data.

For example, data about frequently searched e-waste categories and collection locations could help organizations identify areas where collection infrastructure needs improvement.

Life Cycle Perspective

A product's environmental impact occurs throughout its life cycle, including raw material extraction, manufacturing, transportation, use and end-of-life management. E-Waste Radar primarily addresses the end-of-life stage. Instead of allowing an unwanted device to enter general waste, the platform guides the user towards an appropriate collection channel.

The basic flow is:

Electronic Device → User Decision to Dispose → E-Waste Radar → Suitable Collection Facility → Recycling/Recovery

By improving the connection between users and collection infrastructure, the platform can support better material recovery and reduce inappropriate disposal.

PROPOSED ENGINEERING / IT SOLUTION

Overview of E-Waste Radar

E-Waste Radar is a responsive web application that helps users find appropriate e-waste collection facilities near them. The central idea is simple:

“Tell the system what you want to dispose of and where you are, and it tells you where to take it”

The user does not need to search through multiple websites or manually compare different facilities. The platform can provide:

1. Nearby collection facilities.
2. Distance from the user.
3. Accepted e-waste categories.
4. Opening hours.
5. Contact information.
6. Collection-drive information.
7. Directions through a map.
8. Basic disposal preparation instructions.

Working of the System

The proposed system follows these steps:

Step 1: Select E-Waste Category

The user selects the type of electronic waste.

Step 2: Enter Location

The user can either enter a location manually or allow the website to access their approximate location.

Step 3: Find Suitable Facilities

The system searches its database for facilities that accept the selected category.

Step 4: Calculate Distance

The system calculates the approximate distance between the user's location and available facilities.

Step 5: Display Results

The facilities are displayed in order of relevance, with information like facility name, distance, accepted items, opening hours, current status and contact information.

Step 6: Choose a Facility

The user can open a facility's details and view its location on an interactive map.

Step 7: Disposal Preparation

Before visiting, the user can access a short checklist explaining how to prepare the device.

MAJOR FEATURES OF E-WASTE RADAR

Location-Based Facility Search

The most important feature is location-based searching. Users do not have to know the names of recycling organizations beforehand. They simply provide their location, and the system identifies relevant facilities.

Waste Category Matching

Not every facility necessarily accepts every type of electronic waste. Therefore, the system matches the user's selected waste category with the categories accepted by each facility.

Interactive Map

An interactive map displays the user's approximate location and nearby collection facilities. The user can select a facility to view more information and obtain directions.

Collection Drive Information

Some organizations conduct temporary e-waste collection drives. The platform can display:

- Date
- Location
- Accepted items
- Time
- Registration or booking information

This allows users to participate in collection campaigns even when a permanent facility is not nearby.

Disposal Preparation Guide

The system provides short preparation instructions based on the selected device. This is especially important for devices containing personal information. For example, a laptop disposal checklist may include:

- Back up important files.
- Sign out of accounts.
- Remove external storage devices.
- Delete personal data.
- Follow the facility's instructions for handover.

FEASIBILITY ANALYSIS

Technical Feasibility

The proposed system is technically feasible because it uses commonly available web technologies.

HTML, CSS and JavaScript can be used to create the interface, while databases can store facility information. Map APIs can provide location and navigation functionality.

Therefore, the system can be implemented as a responsive website without requiring specialized hardware.

Economic Feasibility

The basic version can be developed at relatively low cost using open-source technologies.

The major potential expenses would include:

1. Hosting.
2. Database services.
3. Map API usage.
4. Domain registration.
5. Maintenance.

A college-level prototype can be developed using free or low-cost services.

Operational Feasibility

The platform is designed for ordinary users and does not require technical knowledge. A simple interface with clear categories and map-based results makes the system easy to operate. However, facility information must be regularly updated to prevent outdated information from being displayed.

User Feasibility

The system solves a practical problem faced by users who want to dispose of electronic waste responsibly but do not know where to go. The use of familiar web features such as search, maps and filters makes the platform accessible to a wide range of users.

EXPECTED ENVIRONMENTAL BENEFITS

The proposed system can provide several environmental and social benefits.

Improved E-Waste Collection

By making collection facilities easier to find, the system can encourage more users to channel their electronic waste through appropriate collection systems.

Reduction in Improper Disposal

Users who previously discarded electronic devices with ordinary waste may be encouraged to use authorized collection channels.

Better Resource Recovery

Proper collection allows useful materials from electronic products to enter appropriate recycling and recovery processes.

Increased Environmental Awareness

The platform provides users with information about responsible disposal rather than simply giving them a location.

Better Access to Collection Drives

Temporary collection campaigns can reach more users when their information is available through a centralized platform.

Data for Future Planning

Aggregated and privacy-conscious data could help identify areas with high demand for e-waste collection facilities.

JUSTIFICATION OF THE PROPOSED SOLUTION

The main strength of E-Waste Radar is that it addresses the **accessibility problem** in e-waste management. Many digital solutions focus on identifying whether an item can be reused or recycled. However, identifying an item is only one part of responsible disposal. After deciding that a device should be disposed of, the user still needs to know where they should take the said items. E-Waste Radar focuses specifically on this step. The system combines **waste category selection + location + facility information** to provide a practical solution.

It is also different from a reward-based system because users are not encouraged to dispose of e-waste through artificial incentives. Instead, the platform reduces the effort required to make the environmentally responsible choice. The solution is particularly suitable for colleges, residential communities and urban areas where large numbers of electronic devices are used and discarded.

LIMITATIONS AND FUTURE SCOPE

The effectiveness of E-Waste Radar depends heavily on the accuracy of facility information. If a facility changes its operating hours or stops accepting a particular type of e-waste, the database must be updated. Location accuracy may also vary depending on the user's device and mapping service.

Future versions could introduce:

1. AI-based identification of electronic devices.
2. QR codes for collection handover.
3. Verified recycler profiles.
4. Smart collection bins.
5. Notifications for nearby collection drives.
6. Predictive analysis of collection demand.
7. Integration with institutional e-waste programs.
8. Multi-language support.

These improvements could make the platform more useful at a larger scale.

CONCLUSION

Improper disposal of electronic waste is an increasing environmental challenge, and responsible disposal requires both suitable infrastructure and user participation. While collection and recycling facilities may already exist, users often face difficulty in finding the right facility for their particular type of electronic waste.

E-Waste Radar proposes a simple technology-based solution to this accessibility problem. The platform allows users to select their e-waste category and location and then identifies suitable collection facilities based on factors such as distance and accepted waste types. Interactive maps, collection-drive information and disposal preparation guides further improve the usefulness of the system.

The proposed solution supports SDG 12, particularly responsible waste management and environmental awareness, while also contributing to sustainable communities and digital innovation. E-Waste Radar demonstrates how information technology can be used not only to create convenient digital services but also to encourage environmentally responsible behaviour. By making the path from “I have e-waste” to “I know exactly where to take it” simpler, the system can support better participation in formal e-waste management.

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